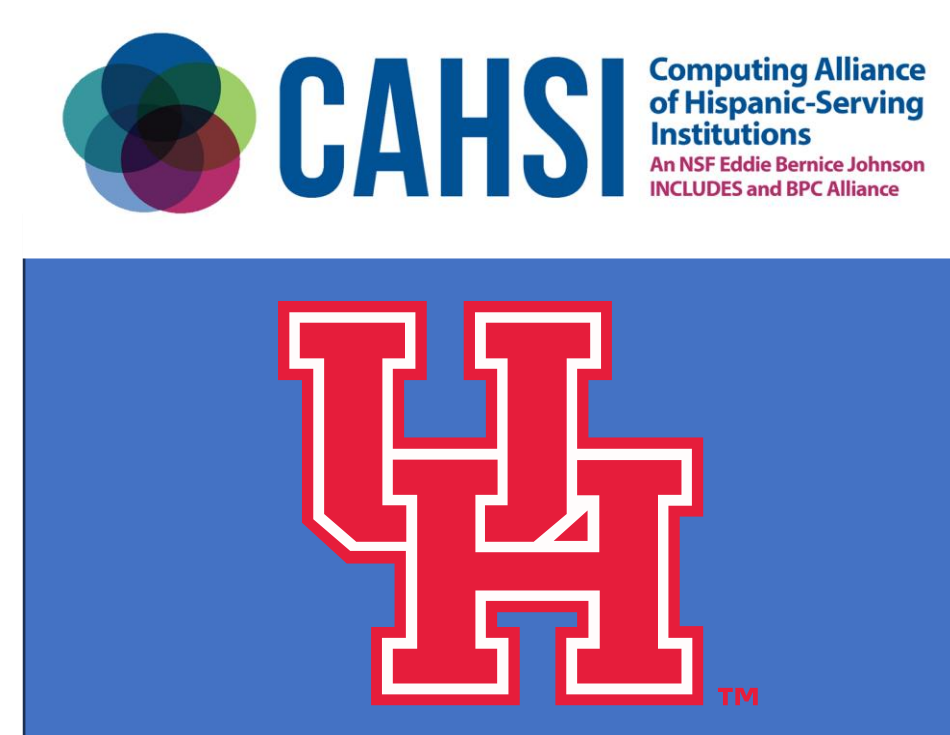




Towards Inclusive Healthcare: Leveraging Low-Resource Neural Machine Translation for Multilingual Communication

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Abstract

Language barriers in healthcare often result in miscommunication, delayed diagnoses, and unequal treatment for speakers of underrepresented languages. Vietnamese, spoken by over 90 million people worldwide and more than 1.5 million in the United States, exemplifies this challenge, as it remains low-resource in natural language processing due to limited bilingual medical data. Neural Machine Translation (NMT) has advanced multilingual communication through transformer architectures, but performance typically favors high-resource pairs. This study investigates whether multilingual large language models (LLMs), specifically MBart50, can address these disparities through domain-specific fine-tuning. Using approximately 200,000 Vietnamese-English parallel sentences from the MedEV medical corpus, MBart50 was fine-tuned and evaluated with BLEU, METEOR, Translation Edit Rate (TER), and BERTScore. Results show significant gains over baseline systems, with BLEU 41.2 and BERTScore 94.4 in Vietnamese-to-English translation. These findings demonstrate the potential of LLMs to deliver accurate medical translations and promote more inclusive, equitable healthcare.

Introduction

Language barriers in healthcare contribute to miscommunication, delayed diagnoses, and worsening health disparities. Speakers of low-resource languages, such as Vietnamese, face particular challenges because existing translation systems are optimized for high-resource language pairs. Neural Machine Translation (NMT) has advanced multilingual communication, but limited parallel data and specialized vocabulary restrict its effectiveness for underrepresented communities. Recent multilingual models like MBart50 provide new opportunities through transfer learning and fine-tuning, allowing knowledge from high-resource languages to benefit low-resource pairs.

Hypothesis: Fine-tuning MBart50 on Vietnamese-English clinical dialogues (MedEV dataset) will significantly improve translation quality and support more inclusive healthcare communication.

Methodology / Approach

- Two-Phase Design:**

 - Phase 1:** Established French-English baselines with ~3k sentence pairs (Global Voices & Europarl) using both traditional and LLM-based models
 - Phase 2:** Fine-tuned MBart50 on ~200k Vietnamese-English medical sentences from the MedEV corpus
- Tools & Evaluation:**

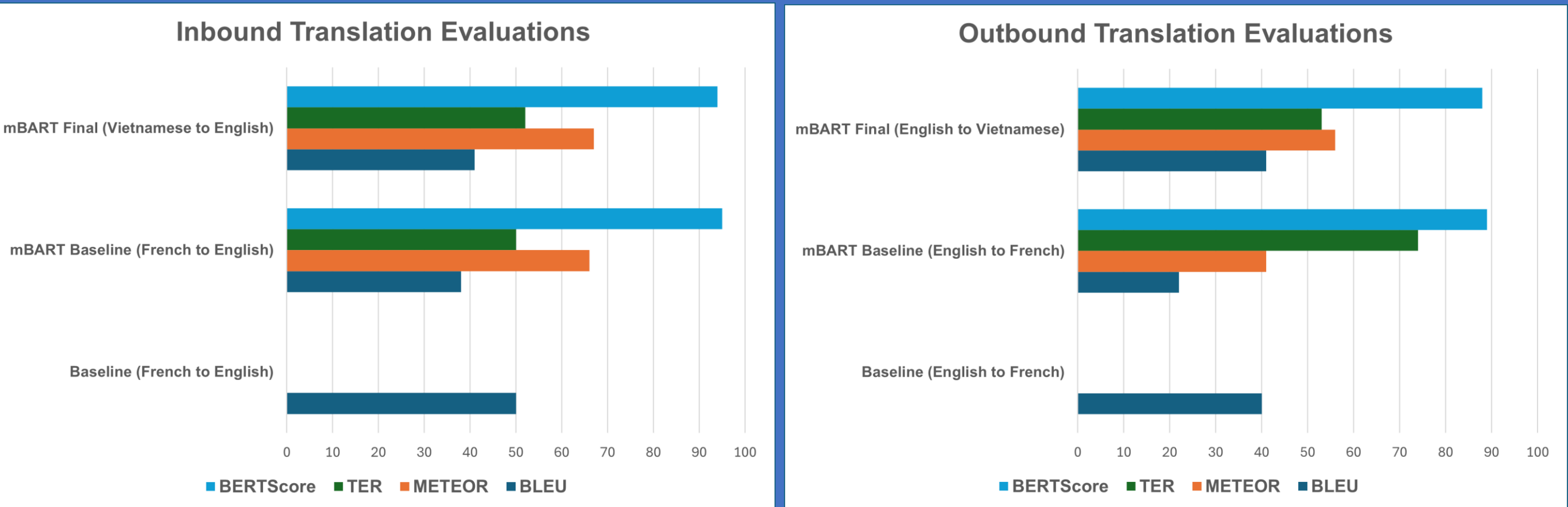
 - Model:** MBart50 (Hugging Face Transformers & Seq2SeqTrainer)
 - Tokenization:** MBart50TokenizerFast, SentencePiece, Moses
 - Evaluation:** BLEU, METEOR, TER, BERTScore
 - Platform:** Google Colab (A100 GPU)
- Limitations:**

 - GPU constraints (batch size, epochs)
 - Vocabulary gaps in Vietnamese medical terms
 - Data scarcity in low-resource settings

Background

Neural Machine Translation (NMT) has advanced multilingual communication through sequence-to-sequence (Seq2Seq) transformer architectures that use self-attention to capture context. These models, however, require large datasets, creating challenges for low-resource languages such as Vietnamese. Pretrained multilingual models like MBart50 help bridge this gap by transferring knowledge from high-resource languages. In this project, MBart50 was fine-tuned on Vietnamese–English medical data using Hugging Face’s Transformers and Seq2SeqTrainer. Translation quality was evaluated with BLEU, METEOR, TER, and BERTScore to measure n-gram precision, semantic alignment, and contextual accuracy. In clinical settings, where precision is critical, effective NMT has the potential to enable real-time communication and improve healthcare for underserved linguistic communities.

Graphs 1 and 2: Evaluation Metrics (BERTScore, TER, METEOR, BLEU) for Inbound and Outbound Trips



Results / Discussion

Translation quality was evaluated with BLEU, METEOR, TER, and BERTScore. The fine-tuned MBart50 Vietnamese-English model outperformed both the baseline and French-English systems, benefiting from domain-specific data and transfer learning.

Inbound (Vietnamese → English): BLEU ~41, METEOR ~67, TER ~52, BERTScore ~94 → strong semantic and contextual alignment with human references.
Outbound (English → Vietnamese): BLEU ~41, METEOR ~56, TER ~52, BERTScore ~88 → slightly lower but still reliable performance.

Compared to French–English baselines (BLEU ~50 inbound, ~40 outbound), the Vietnamese model shows that targeted fine-tuning and LLM integration can push low-resource languages toward competitive translation quality. Evaluation graphs further illustrate that MBart50 with medical-domain fine-tuning significantly outperforms traditional NMT baselines.

Future Work

To build on these results, future research should:

- Extend to other low-resource medical languages
- Integrate spoken dialogue and clinical data for better real-world performance
- Apply domain-adaptive pretraining or instruction tuning for medical relevance
- Develop lightweight, mobile-friendly models
- Conduct user-centered evaluations with healthcare professionals and native speakers

These directions will improve the practicality and accessibility of neural machine translation in multilingual healthcare settings.

Summary / Conclusion

This research demonstrates that multilingual pretrained models like MBart50 can bridge healthcare communication gaps for underrepresented languages. Fine-tuning on domain-specific Vietnamese-English medical data yielded strong translation quality, outperforming traditional baselines. These results confirm that LLMs, combined with domain-targeted training, can deliver inclusive and reliable NMT for critical fields like healthcare.

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